HDOC DAY 62

PROBLEM

Write a C program that reads the sizes and elements of two sorted variable-length integer arrays, arr1[] and arr2[], containing their elements in ascending order. Pass both arrays to a function named merge_sorted_arrays() using their addresses. Use pointers to pass the arrays between the calling function and merge_sorted_arrays(). The function merge_sorted_arrays() must:

- Merge arr1[] and arr2[] into a third array, merged[].
- Maintain the ascending order of the elements in merged[].
- Preserve duplicate elements, if any.
- Display the elements of merged[] separated by spaces.
- Use the two-pointer technique to merge the arrays efficiently.

CODE

```
#include <stdio.h>


void merge_sorted_arrays(int *arr1, int n1, int *arr2, int n2, int *merged)
{
   int i = 0, j = 0, k = 0;


   /* Merge while both arrays have elements */
   while (i < n1 && j < n2) {
      if (*(arr1 + i) <= *(arr2 + j))
         *(merged + k++) = *(arr1 + i++);
      else
         *(merged + k++) = *(arr2 + j++);
   }
```

```
    /* Copy remaining elements of arr1 */
    while (i < n1)
        *(merged + k++) = *(arr1 + i++);

    /* Copy remaining elements of arr2 */
    while (j < n2)
        *(merged + k++) = *(arr2 + j++);

    /* Display merged array */
    for (i = 0; i < k; i++) {
        printf("%d", *(merged + i));

        if (i < k - 1)
            printf(" ");
    }

    printf("\n");
}

int main(void)
{
    int n1, n2;

    scanf("%d", &n1);
    int arr1[n1];

    for (int i = 0; i < n1; i++)
        scanf("%d", &arr1[i]);
```

```
    scanf("%d", &n2);
    int arr2[n2];

    for (int i = 0; i < n2; i++)
        scanf("%d", &arr2[i]);

    int merged[n1 + n2];

    merge_sorted_arrays(arr1, n1, arr2, n2, merged);

    return 0;
}
```